



Mikhail Vasiliev

Deep Learning Engineer

Experience

2025—
present **Senior Machine Learning Engineer, Raiffeisenbank**

2023—2025 **Senior Machine Learning Specialist, Makves**

Project: *Development and implementation of RAG system*

- Developed and implemented a RAG system for automating customer request processing
- Optimized system hyperparameters using Ragas library and GigaChat LLM

Tools: LangChain, Ollama, Saiga, GigaChat, HuggingFace, FastAPI, Ragas

Project: *Creation of a comprehensive security solution for corporate networks based on unstructured data*

- Implemented a neural network module for detecting violations of personal data laws, increasing detectable classes from 14 to 36 with top 1 accuracy reaching 98.9
- Developed a module for analyzing scanned document content: text, tables, stamps, signatures and corporate forms detection, increasing classes from 5 to 19 with mAP@.5 improved from .89 to .94
- Implemented sensitive data detection in text files with added NER module
- Created an ensemble of algorithms for anomaly detection in tabular data, including time series
- Developed sensitive data detection in audio files
- Collected and organized labeling for 8 datasets for classification and object detection tasks

Tools: transformers, YOLO, PyOD, Pandas, Sklearn, PyTorch, lightning, NumPy, HuggingFace, ONNX, FastAPI, uvicorn, PyInstaller, optimum, CatBoost, CVAT, natasha

Personal Projects

2024 **Team Lead and Technical Expert**, *CheckDocAI*

Project: *Telegram bot with AI module for document quality control for Gulfstream LLC, significantly reducing verification time and improving accuracy*

- Led a team of two data scientists and a backend developer, responsible for project development and implementation
- Successfully deployed for commercial use with monthly savings of 40 man-hours

Tools: aiogram, YOLO, ONNX, Albumentations, CVAT

Hackathons

2024 **VK HSE Data Hack**, *1st place*

Hackathon for news article classification into 21 categories. Our solution combined results from a small transformer-based classifier and LLM predictions

- Enriched the dataset
- Selected zero-shot classification model
- Trained classifier model
- Coordinated team work
- Presented results

Tools: transformers, Saiga3 8b, taiga dataset, streamlit

Talks

2026 **Anatomy of Modern Transformers**, *Data Fest*

2025 **Anomaly Detection with Python**, *Positive Hack Days*

2024—2025 **NLP and CV Neural Networks in Data Protection**, *Moscow Python Meetup*

Education

- 2024 **Data Analysis with SQL**, Training Center “Specialist”, professional development
- 2022—2023 **Computer Vision Engineer**, Deep Learning School, MIPT, professional retraining
- 2022 **Data Science Specialist**, Yandex Practicum, professional retraining
- 2021—2022 **Introduction to AI and Neural Networks for Aviation Applications**, MAI, professional development
- 2005—2008 **Translation and Translation Studies**, MAI, specialist degree
- 2003—2009 **Aviation and Space Thermal Engineering**, MAI, specialist degree

Languages

Russian	■■■■■	native
English	■■■■■	B2
German	■■■■■	B2
Esperanto	■■■■■	B2

Skills and Technologies

Deep Learning & LLMs

- RAG, Prompt Engineering
- Qwen, Llama, GigaChat
- LangChain, Ollama, Ragas
- Transformers, BERT

Computer Vision

- YOLOv8, U-Net
- OpenCV, CVAT
- Albumentations

Anomaly Detection

- PyOD, RRCF
- Isolation Forest, ECOD
- HBOS, PySAD

NLP & Speech

- NER (natasha)
- Whisper, HuBERT
- Text classification

Vector Search

- FAISS, Qdrant, Milvus
- sentence-transformers

MLOps & Deployment

- Docker, Linux, FastAPI
- MLflow, Airflow
- ONNX, PyInstaller

Testing & Reproducibility

- pytest
- Git, CI/CD basics
- Experiment tracking

Data & Core ML

- Python, SQL, pandas
- scikit-learn, CatBoost
- PostgreSQL, MySQL